K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to University of Mumbai

A Project Based Learning

Report On

“Comparative Time-Series Forecasting of Food Inflation using El Niño Indices: ARIMA vs. Prophet vs. LSTM”

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CERTIFICATE

This is to certify that the project titled “Comparative Time-Series Forecasting of Food Inflation using El Niño Indices: ARIMA vs. Prophet vs. LSTM” is completed under my supervision and guidance in partial fulfilment of the requirements of the course Project Based Learning - Minor Project Lab-1, by the following students:

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The course is a part of Semester V of the Department of Artificial Intelligence and Data Science during the academic year 2026-2027. The said work has been assessed and is found to be satisfactory.

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College Seal

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CHAPTER 1 | INTRODUCTION

1.1 Background

Food inflation refers to changes in the prices of food items over time. In India, food price fluctuations matter for household budgets, purchasing power, supply planning and inflation monitoring. The Consumer Food Price Index (CFPI) is an official indicator that tracks changes in retail prices of food items consumed by households [1].

Food prices are influenced by several interacting drivers, including agricultural output, rainfall distribution, extreme weather, transportation costs, seasonal availability and market conditions. El Niño is one phase of the El Niño-Southern Oscillation (ENSO), a climate phenomenon associated with changes in tropical Pacific sea-surface temperatures. The Oceanic Niño Index (ONI), published by NOAA, summarizes sea-surface temperature anomalies in the Niño 3.4 region [2]. Climate variability can influence food-supply conditions, but its predictive value for a specific inflation series requires empirical testing.

1.2 Problem Statement

Traditional inflation forecasting may rely on historical inflation values alone. A research question is whether an additional climate-related signal, ONI, improves the prediction of Indian food inflation. A model can become more complex after adding external predictors without necessarily becoming more accurate. Therefore, an objective comparison against a historical-only baseline is necessary.

1.3 Proposed Approach

The project builds a monthly food-inflation dataset, aligns it with ONI, develops ARIMAX, Prophet and LSTM forecasting approaches, and evaluates their predictions on a common January 2024–December 2025 holdout period. A univariate LSTM baseline is included to isolate the practical benefit of the ONI feature. The evaluated outputs are presented through a FastAPI service and a React dashboard.

1.4 Scope and Boundaries

- Geographical focus: India, using the food-inflation series available in the project dataset.
- Time coverage: January 2012 to December 2025; 168 monthly observations.
- Primary variables: food inflation (%) and ONI; the month field supports time indexing.
- Model benchmark: ARIMAX, Prophet, LSTM + ONI and a historical-inflation-only LSTM baseline.
- Output: documented model comparison, stored test predictions, REST API endpoints and visualization dashboard.

This is a research and evaluation prototype. The deployed API serves saved predictions and metrics; it is not a continuously retrained forecasting service and should not be described as an operational government forecasting system.

CHAPTER 2 | AIM & OBJECTIVE

2.1 Aim

To investigate the usefulness of El Niño-related climate information for forecasting monthly Indian food inflation by implementing and comparing statistical and deep-learning time-series models, and to communicate the evaluation results through a web-based dashboard.

2.2 Objectives

1. Collect and organize historical CPI/CFPI-derived food inflation observations and NOAA ONI data.
2. Clean dates, address missing values where supported by the source index, and merge the monthly series.
3. Examine time-series properties and construct appropriate lagged ONI features for statistical models.
4. Implement ARIMAX and Prophet models incorporating climate information.
5. Implement a multivariate LSTM using food inflation and ONI, alongside a univariate LSTM baseline.
6. Assess model behaviour on a fixed, chronological 24-month test window using MAE, RMSE, sMAPE and MASE.
7. Compare relative errors, identify the strongest observed method, and discuss limitations of the evidence.
8. Publish processed results through a FastAPI backend and an interactive web dashboard.

2.3 Deliverables and Success Criteria

Deliverable	Verification basis
Aligned dataset	Processed cpi_oni.csv contains dated food-inflation and ONI values.
Model outputs	Four prediction CSVs with actual and predicted values for the test period.
Comparison results	model_comparison.csv includes all four accuracy measures and ranks.
Backend	Documented routes for data, prediction, comparison and latest stored forecast.
Frontend	Sections for overview, model predictions, evaluation and CSV metric export.

Success is defined as completing a reproducible, transparent comparison and correctly presenting its findings. Success does not require the ONI-augmented model to rank first.

CHAPTER 3 | LITERATURE SURVEY

3.1 Relevant Literature and Sources

The literature review combines authoritative data sources, foundational forecasting methods, evaluation research and recent evidence on climate-related food price risks. The entries below provide methodological support rather than independent validation of this project's numerical results.

Source	Key contribution	Relevance to project
[1] MoSPI, CFPI / CPI	Defines official food price indices and monthly CPI release practices.	Supports selection and interpretation of the dependent variable.
[2] NOAA ONI	Defines an ENSO-associated index constructed from Niño 3.4 sea-surface temperature anomalies.	Supports the external climate feature.
[3] Hochreiter & Schmidhuber (1997)	Introduces Long Short-Term Memory networks for learning sequence dependencies.	Foundation for both project LSTM variants.
[4] Taylor & Letham (2018)	Develops a configurable trend/seasonality-based forecasting method.	Basis of the Prophet experiment.
[5] Hyndman & Koehler (2006)	Studies limitations of error measures and proposes MASE.	Justifies comparing MAE, RMSE, sMAPE and MASE with care.
[6] Climate-shock study (2026)	Examines climate shocks and food-price inflation in India.	Motivates climate-aware research without claiming a direct causal ONI effect.
[7] Statsmodels SARIMAX	Documents ARIMA-family models with exogenous regressors.	Implementation basis for the ARIMAX benchmark.

3.2 Identified Research Gap

Climate events provide a plausible supply-side pathway to food-price changes, but this does not establish that ONI will improve a particular forecasting algorithm. A focused, matched-dataset experiment is needed to test whether using ONI improves predictions relative to a model trained on the inflation history alone.

3.3 Project Contribution

This work contributes a student-developed comparison across two statistical methods and two LSTM variants, a consistent test calendar, four quantitative evaluation metrics, and a dashboard that makes the results inspectable. Its key contribution is the explicit baseline comparison: the outcome is reported even though the ONI-augmented LSTM does not win.

CHAPTER 4 | SOCIAL IMPACT/EFFECT OF PROBLEM

4.1 Why the Problem Matters

Unexpected food-price changes can reduce household purchasing power and complicate decisions made by farmers, traders, distributors and policy analysts. Understanding possible predictive signals is relevant to long-term work on food security and inflation analysis. Recent published research connects certain climate shocks with changes in Indian food-price inflation, while also emphasizing that effects vary by shock and time horizon [6].

4.2 Potential Beneficiaries

- Households and researchers: greater awareness of how food inflation has fluctuated over time.
- Agricultural and supply-chain researchers: a starting point for studying weather-to-price transmission.
- Academic users: transparent comparisons of baseline and climate-informed time-series approaches.
- Policy analysts: an exploratory example of why external indicators should be tested rather than assumed to improve prediction.

4.3 Responsible Interpretation

The prototype cannot support decisions about food procurement, monetary policy or commodity trading without additional independent validation. Forecast errors may be large during unusual periods, and a single national series cannot represent differences among Indian states or food categories. ONI is an indicator, not a direct measurement of local crop loss or domestic prices.

4.4 Positive Outcome Even When the Hypothesis Is Not Supported

A negative experimental result is useful: if historical inflation alone provides lower prediction error, adopting a more complex climate-augmented model is not justified by this benchmark. Publishing that result supports evidence-based model selection and reduces the risk of misleading claims about AI or climate effects.

CHAPTER 5 | SYSTEM DESIGN

5.1 High-Level Architecture

The design separates offline research workflows from online presentation. Notebooks read source files, prepare a combined dataset, train and evaluate candidate models, and write prediction/metric CSV files. The Python backend loads those stored CSVs and exposes JSON endpoints. The React client requests the data and renders charts and scorecards. There is no trained-model inference call inside the current API request path.

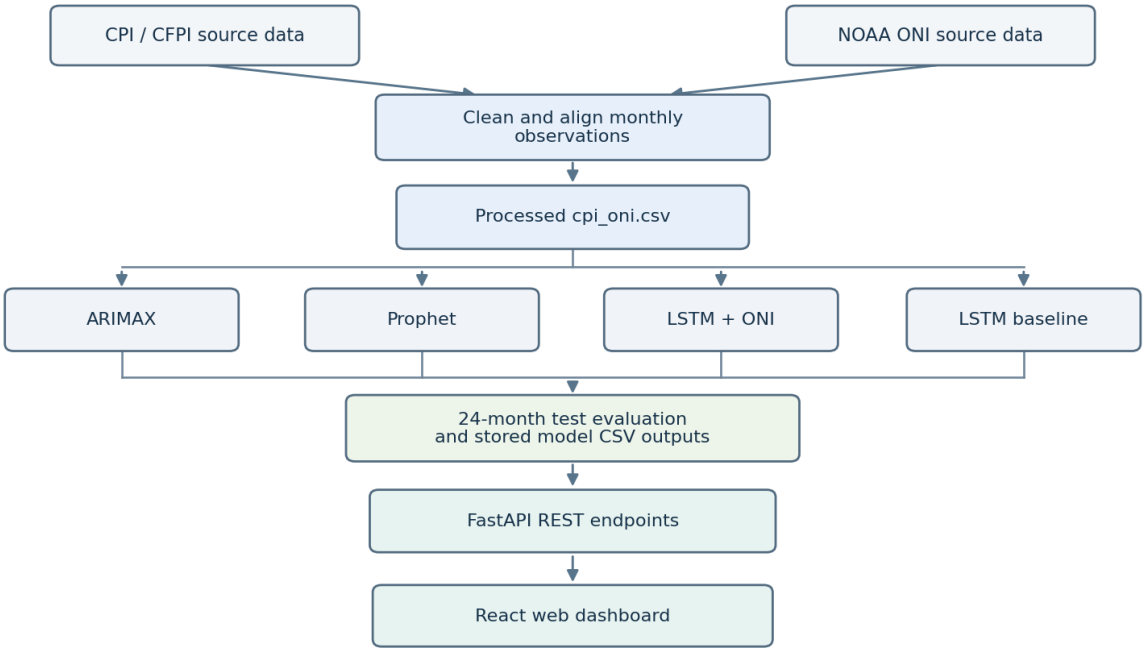


Figure 5.1. Implemented project-level data, modeling and presentation architecture.

5.2 Data Entities

Field	Type / Unit	Role
date	Monthly date (YYYY-MM-DD)	Joins data and identifies observations.
food_inflation	Numeric (%)	Dependent / forecast target.
oni	Numeric index	Climate-related external predictor.
month	Integer (1–12)	Calendar-month identifier.
predicted_food_inflation	Numeric (%)	Stored model prediction.
actual_food_inflation	Numeric (%)	Test-period reference value.

5.3 Functional Modules

Module A (data preparation) converts CPI source records to monthly dates, derives inflation from year-over-year index changes for missing source values where available, standardizes ONI dates, handles missing-value codes, and merges on date. Module B (modeling) fits the four selected forecasting methods. Module C (evaluation) computes quantitative metrics and rankings. Module D (API) returns the saved result sets. Module E (dashboard) visualizes the model outputs.

5.4 API Interface Design

Endpoint	Purpose
GET /data	Returns processed time-series observations.
GET /predictions	Returns stored predictions; supports an optional model filter.
GET /comparison	Returns MAE, RMSE, sMAPE and MASE for all four ranked models.
GET /forecast	Returns the latest dated row saved for each model.
GET /forecast/{model_name}	Returns the latest dated row saved for the requested model.
GET /health	Reports basic service health.

In this API, “latest forecast” means the final record currently present in the saved prediction CSV. Since the checked-in results end in December 2025, this endpoint should not be represented as a real-time future forecast for November 2026.

5.5 Non-Functional Considerations

- Reproducibility: source notebooks, processed files and model-comparison outputs are versioned in the repository.
- Clarity: response field names, model identifiers and error metrics are explicit.
- Reliability: the dashboard includes saved metric fallback values when API calls fail; historical forecast lines still depend on successful data retrieval.
- Extensibility: new model CSV outputs and backend model mappings can be added without redesigning all interfaces.

CHAPTER 6 | FLOW CHART

Figure 6.1 follows the implemented analytical workflow, from source acquisition to comparison and web presentation. The central evaluation step is separated from deployment: training occurs in notebooks, while the deployed application serves previously generated results.

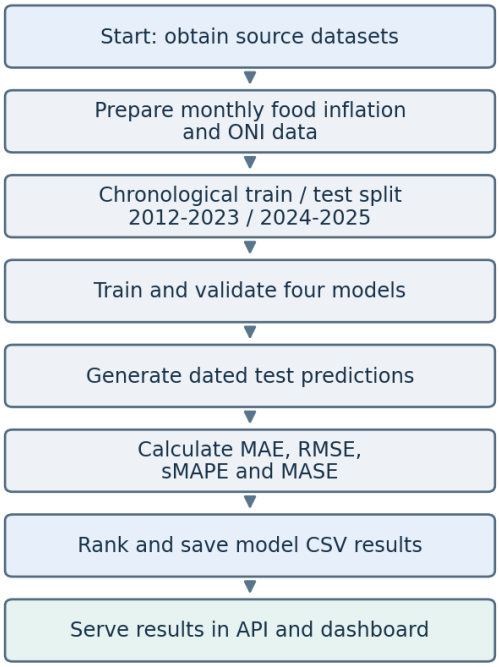


Figure 6.1. End-to-end project flowchart based on the repository structure.

6.1 Description of Processing Steps

1. Source input: read the CPI data spreadsheet and NOAA ONI time-series file.
2. Preprocessing: standardize dates, fill eligible missing inflation values, join by month, and verify continuity.
3. Time separation: retain January 2024 through December 2025 as a 24-month test period.
4. Model development: fit statistical models and train both LSTM configurations on historical data.
5. Test prediction: create dated output records and retain actual values for evaluation.
6. Evaluation: calculate errors, assign rank for each metric, and compute average ranks.
7. Presentation: expose CSV results via FastAPI and chart them in the React dashboard.

6.2 Decision Rules

Model selection is empirical: a lower MAE, RMSE, sMAPE or MASE denotes better performance for the reported evaluation. The average rank is computed from the four individual metric ranks. In the current experiment the same method ranks first for all four metrics, so there is no tie-resolution ambiguity.

CHAPTER 7 | SYSTEM IMPLEMENTATION

7.1 Software and File Organization

Layer	Implemented technologies / artifacts
Data preparation	Python, Pandas, NumPy, Excel/CSV source files and Jupyter notebooks.
Time-series models	Statsmodels SARIMAX (ARIMAX); Prophet; TensorFlow/Keras LSTMs.
Evaluation	Scikit-learn / NumPy calculations; comparison CSVs and figures.
REST backend	FastAPI, Pydantic, Uvicorn; routes in backend/routes/.
User interface	React, Vite, Recharts, Lucide, Express-based server-side rendering.
Deployment	Frontend URL on Vercel; API URL on Render (as recorded in source configuration).

7.2 Data Preprocessing

The source files are data/raw/cpi.xlsx and data/raw/oni.csv. The preprocessing notebook creates date values from the source year and month fields. For missing food-inflation rates, it computes a 12-month percentage change from the corresponding price index within the source grouping. It standardizes the ONI date format and replaces NOAA missing-value sentinels with nulls before merging by month.

The final processed file data/processed/cpi_oni.csv contains 168 monthly observations, from January 2012 through December 2025, with columns date, food_inflation, oni and month. The common chronological split uses January 2012–December 2023 (144 months) for the training-side history and January 2024–December 2025 (24 months) for the test period. Applying the three-month lagged ONI regressor removes the first three training-calendar months for those model fits.

The year-on-year calculation used for an eligible missing inflation value is:

$$Inflation(t) = [(PriceIndex(t) / PriceIndex(t-12)) - 1] \times 100$$

This equation is a preprocessing method, not a guarantee that every originally missing record is recoverable; missingness and index comparability must be checked against the underlying source [1].

7.3 ARIMAX Model

The final statistical ARIMAX experiment uses Statsmodels SARIMAX with non-seasonal order (0,1,3), a constant trend and a lagged three-month ONI average as the exogenous regressor. Specifically, ONI_MA3(t) = mean[ONI(t-1), ONI(t-2), ONI(t-3)]. This uses past months for each climate feature. The model produces a multi-step forecast across the 24-month holdout with the corresponding exogenous-regressor sequence provided [7]. The approved project title uses the broader term ARIMA; the implemented statistical model is its exogenous-variable variant, ARIMAX.

7.4 Prophet Model

Prophet models trend and annual seasonality with additive ONI_MA3 regression. The final notebook uses yearly seasonality enabled, additive seasonality, changepoint_prior_scale = 0.10, seasonality_prior_scale = 1.0, and a 95% prediction interval setting. The prediction CSV stores the point estimate and its lower and upper bounds [4].

7.5 LSTM + ONI Model

The multivariate LSTM uses two input features (historical food inflation and ONI), scaled with a MinMaxScaler fitted to training data only [8]. The selected final input sequence contains the preceding 18 months. Its network has an LSTM layer with 64 units, Dropout(0.2), Dense(32, ReLU) and Dense(1); it is trained using the Adam optimizer and mean squared error loss. Training-side validation and early stopping are used to select the final training duration [3], [9].

7.6 LSTM Baseline

The baseline uses only lagged historical food inflation and a 12-month input sequence. Its network uses LSTM(64), Dropout(0.2), Dense(32, ReLU) and Dense(1), and the same general Adam/MSE training setup. Keeping this baseline is essential for evaluating whether the climate feature contributes useful incremental predictive information.

7.7 Metrics and Ranking

For actual y(t), prediction p(t), test length n and training values z(t), the implementation uses the following four measures:

$$MAE = (1/n) \times \Sigma |y(t) - p(t)|$$

$$RMSE = \mathrm{sqrt}\{(1/n) \times \Sigma [y(t) - p(t)]^2\}$$

$$sMAPE\ (\%) = (100/n) \times \Sigma [2|y(t) - p(t)| / (|y(t)| + |p(t)| + \varepsilon)]$$

$MASE = MAE / \mathrm{mean}(|z(t) - z(t-1)|)$, calculated over consecutive training observations.

The sMAPE denominator contains a small epsilon (1e-8) in the implementation. Its percentage values can still be unstable or large when actual and predicted inflation rates are close to zero. The ranking notebook assigns ranks for each metric and averages the four ranks [5].

7.8 Serving and Visualization

The backend uses FastAPI route modules data.py, predictions.py and comparison.py. It reads stored CSVs at request time, performs date and numeric coercion for prediction endpoints, and returns JSON. The frontend uses React and Recharts, provides selectable model lines and evaluation metrics, can export the scorecard as CSV, and distinguishes connected API results from saved comparison fallback values.

7.9 Reproducibility and Deployment Limitations

The repository supplies notebooks and generated prediction CSVs, but the deployed API does not load saved LSTM weight files or retrain models on demand. Consequently, reproducing a model result from scratch requires a compatible Python environment and rerunning the notebook workflow. Neural-network results may vary between training runs due to initialization and runtime differences unless the complete environment and random-state controls are preserved.

CHAPTER 8 | RESULTS AND ANALYSIS

8.1 Experimental Configuration

Property	Observed configuration
Processed dataset	168 monthly records, January 2012–December 2025.
Training-side history	January 2012–December 2023 (144 calendar months).
Holdout window	January 2024–December 2025 (24 calendar months).
Models in final ranking	LSTM baseline, LSTM + ONI, ARIMAX and Prophet.
Comparison measures	MAE, RMSE, sMAPE and MASE (lower is better).

8.2 Model Comparison Table

Model	MAE	RMSE	sMAPE (%)	MASE	Rank
LSTM Baseline	1.3335	1.6460	43.8900	0.9910	1
LSTM + ONI	1.5245	1.9422	49.6371	1.1330	2
ARIMAX	4.6550	6.3055	85.4437	3.4594	3
Prophet	5.0169	6.6850	87.4038	3.7284	4

Table 8.1. Repository-reported final values, displayed to four decimal places (rounded for the report). The full-precision metric values remain available in data/processed/model_comparison.csv.

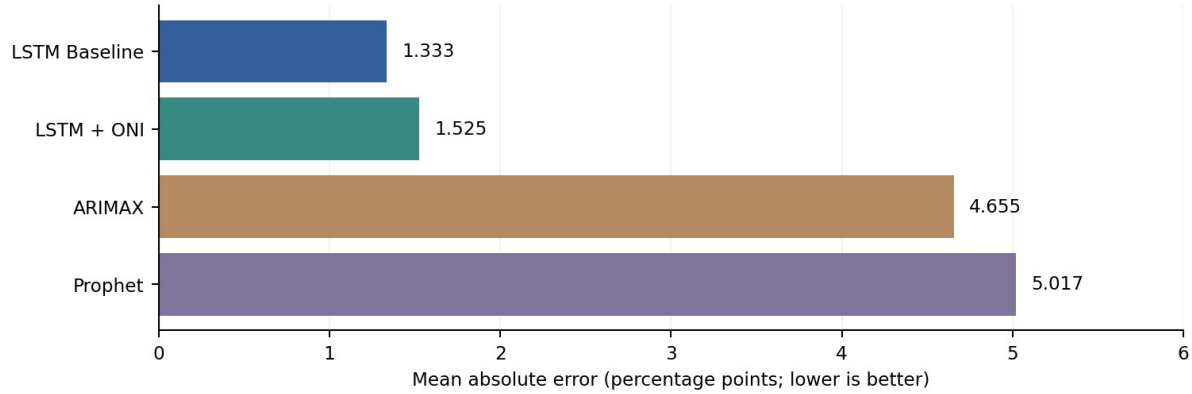


Figure 8.1. MAE comparison for the four benchmark models (source: model_comparison.csv).

8.3 Analysis of the LSTM Models

The baseline LSTM has the lowest error on every reported measure. Its MAE is 1.3335 percentage points, while the ONI-augmented LSTM obtains 1.5245 percentage points. The latter is approximately 14.3% higher relative to the baseline MAE. The result answers the core research question for this configuration and test period: ONI did not improve the best observed LSTM performance.

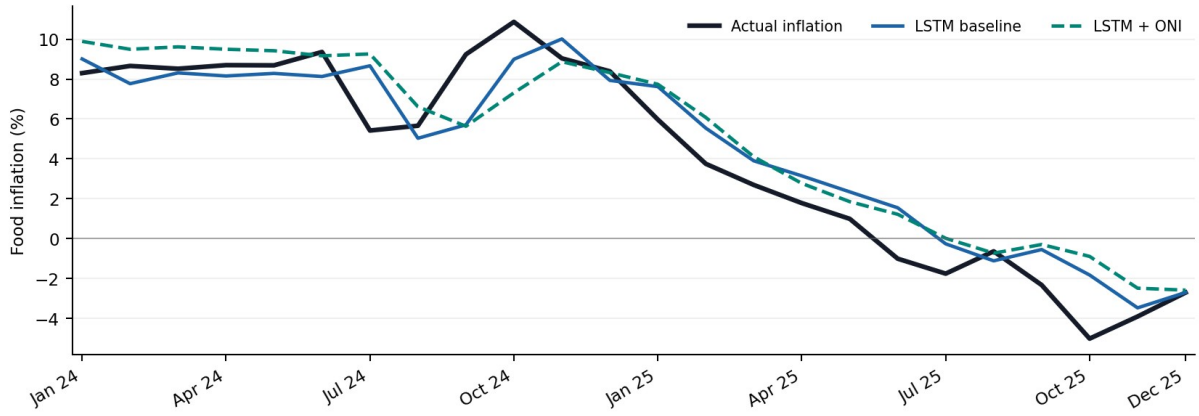


Figure 8.2. Actual food inflation versus the two LSTM test-period prediction sequences (Jan 2024–Dec 2025).

Both LSTM prediction sequences respond to the decline in observed food inflation over the later part of the holdout period more closely than the statistical models. The result should not be interpreted as a general claim that the baseline will always outperform ONI-augmented models on other periods, horizons or economic regimes.

8.4 Analysis of ARIMAX and Prophet

ARIMAX and Prophet rank third and fourth respectively. Their stored prediction trajectories remain positive over much of 2025, whereas the observed food-inflation series falls below zero during parts of that year. Consequently, the longer-horizon point errors increase in the declining-inflation period.

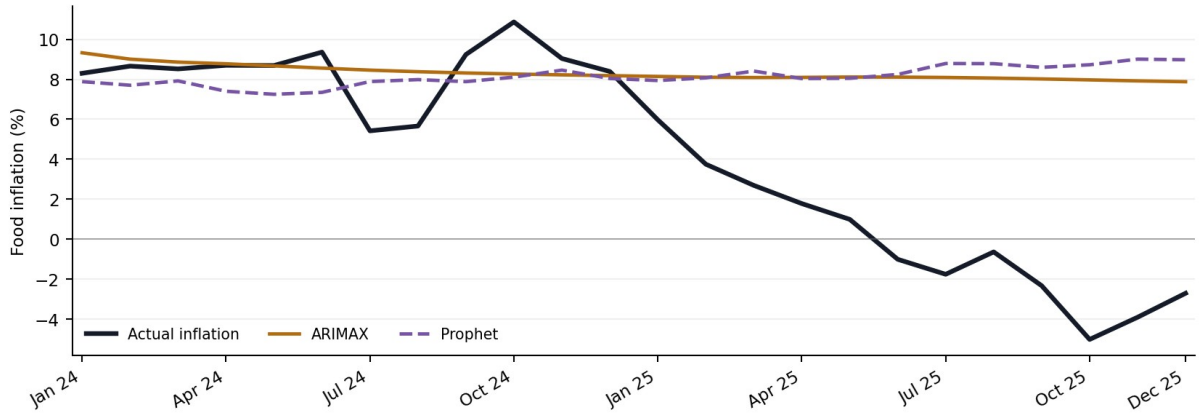


Figure 8.3. Actual food inflation versus ARIMAX and Prophet predictions on the fixed test period.

8.5 Interpreting the Measures

MAE expresses average absolute misses in percentage points because the target itself is an inflation rate. RMSE gives greater weight to large errors and confirms the same model ordering. MASE values below 1 indicate lower error than the in-sample naive-change scale used in the notebook, not guaranteed superiority to every possible out-of-sample naive forecast. sMAPE values are comparatively high and must be interpreted cautiously when observed inflation approaches zero or turns negative [5].

8.6 Important Evaluation Limitations

The four models share a common test date range, but their information conditions are not fully identical. The LSTM test-sequence construction uses the preceding observed test-period inflation values after the first holdout month; this is closer to a rolling, one-step-ahead prediction exercise than a single origin 24-month forecast. By contrast, ARIMAX produces a multi-step forecast, while ARIMAX and Prophet receive test-period lagged ONI regressor values. The comparison therefore indicates observed performance under the implemented procedures, not a definitive like-for-like ranking at a single common forecast horizon.

For a stricter future study, each model should be evaluated under the same origin, horizon, feature-availability policy and rolling-origin protocol. In particular, all external regressors must be available or forecast at the stated decision time. Model hyperparameters should be selected using training/validation data only, with the final test period kept outside tuning.

8.7 Key Findings

- The historical-only LSTM obtained the lowest error across all four measures.
- The LSTM with ONI ranked second and did not surpass the baseline.
- ARIMAX and Prophet produced substantially larger errors in the 2024–2025 period.
- The dashboard successfully represents stored model outputs and comparative evaluation, but not real-time model retraining.
- The measured result applies to this dataset, features, training procedure and evaluation window; it does not prove the absence of a climate-inflation relationship.

CHAPTER 9 | CONCLUSION

9.1 Conclusion

This Project Based Learning work implemented a comparative time-series forecasting system for Indian food inflation with climate-related ONI information. Raw CPI and ONI sources were processed into a dated monthly dataset, four selected models were evaluated on a fixed January 2024–December 2025 test window, and the output was presented through a FastAPI-based service and a React dashboard.

The best reported method was the univariate LSTM baseline (MAE 1.3335, RMSE 1.6460, sMAPE 43.8900% and MASE 0.9910). The LSTM with ONI achieved second place. Thus, the experiment did not show an accuracy advantage from ONI over the historical-only LSTM, despite the theoretical relevance of climate variability to agricultural supply and food prices.

The project demonstrates the importance of transparent baselines, chronological evaluation and balanced reporting. The findings are exploratory and depend on modeling assumptions, variable availability and forecast horizon; they should not be presented as proof of causation or as operational forecasts for public decision-making.

9.2 Scope for Future Improvements

- Use rolling-origin evaluation with matched forecast horizons for all competing models.
- Compare against simple seasonal naive and autoregressive benchmarks in addition to LSTM.
- Test alternative ONI lag structures, rolling summaries and interactions with rainfall, monsoon anomalies or agricultural indicators.
- Assess state-level or food-category-specific inflation data when consistently available.
- Record model versions and environments, automate retraining where appropriate, and expose genuine future-date forecasting only after validation.
- Add residual diagnostics, uncertainty evaluation, model explainability and sensitivity analysis to establish more reliable conclusions.

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